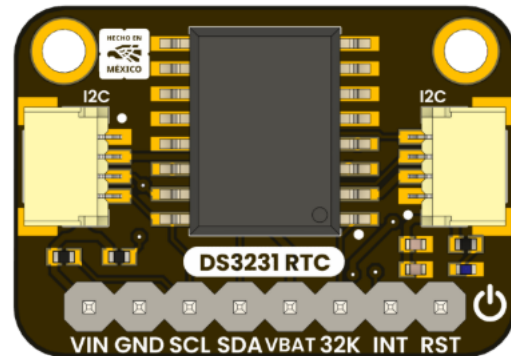


Product Reference Manual (V1.0)

Description

The **DevLab I²C DS3231 RTC Module** provides an exceptionally accurate and reliable solution for real-time clock applications in embedded systems. By integrating a **temperature-compensated crystal oscillator (TCXO)** and an internal crystal within a single package, the DS3231 minimizes timing errors caused by temperature fluctuations, ensuring long-term stability without the need for external calibration. The module automatically adjusts for months with fewer than 31 days and accounts for leap years, maintaining precise time data for decades. Its ability to switch seamlessly to a backup battery supply allows uninterrupted operation even during main power failures, making it ideal for systems that require continuous timekeeping.

Designed for easy integration, the module uses a **standard I²C communication interface** compatible with most microcontrollers, supporting both 3.3 V and 5 V logic levels. Developers can configure two programmable alarms, a square-wave output, and an optional reset function for added system control. With its low power consumption, compact design, and broad temperature tolerance, the DS3231 RTC Module is well suited for **data loggers, IoT nodes, industrial controllers, wearable devices, and consumer electronics** that demand dependable, autonomous time management in any environment.



Key Features

- **High Accuracy RTC:** Maintains timekeeping with ± 2 ppm from 0 °C to +40 °C and ± 3.5 ppm from -40 °C to +85 °C.
- **Integrated TCXO and Crystal:** Provides temperature-compensated stability without external components.
- **Battery Backup Support:** Keeps clock operation uninterrupted during power loss.
- **Dual Alarms and Square-Wave Output:** Two independent alarms and a configurable INT/SQW output.
- **I²C Interface:** Standard two-wire serial communication up to 400 kHz for simple integration.
- **Wide Voltage Compatibility:** Operates with 3.3 V or 5 V logic systems.
- **Wide Temperature Range:** Reliable operation across -40 °C to +85 °C for industrial and commercial use.
- **Compact DevLab Form Factor:** Includes a **QWIIIC-compatible JST 1.0 mm connector** for quick and secure I²C expansion.

DevLab format compatibility.

Simplicity and compatibility are the core principles of the **DevLab form factor**. This standard defines a **compact and communication-optimized board layout**, ensuring straightforward connection and interoperability among DevLab modules.

Software Support

- **Arduino IDE**
- **Micro Python**

Applications

- **Embedded Systems:** Provides a precise time reference for microcontroller-based designs and real-time operations.
- **Data Logging:** Enables accurate timestamping of sensor readings or recorded measurements.
- **IoT Devices:** Maintains synchronization and scheduling for automation, communication, and energy management tasks.
- **Wearable Technology:** Delivers dependable timekeeping in compact, low-power devices.
- **Consumer Electronics:** Used in clocks, cameras, GPS trackers, and smart home controllers.
- **Industrial and Utility Equipment:** Ensures reliable timing for power meters, control panels, and monitoring systems.
- **Server and Network Systems:** Offers stable time synchronization for data storage and event tracking.
- Haptic user interfaces and research platforms

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1 The Board

1.1 Accessories

- 1×6-pin 2.54 mm male header

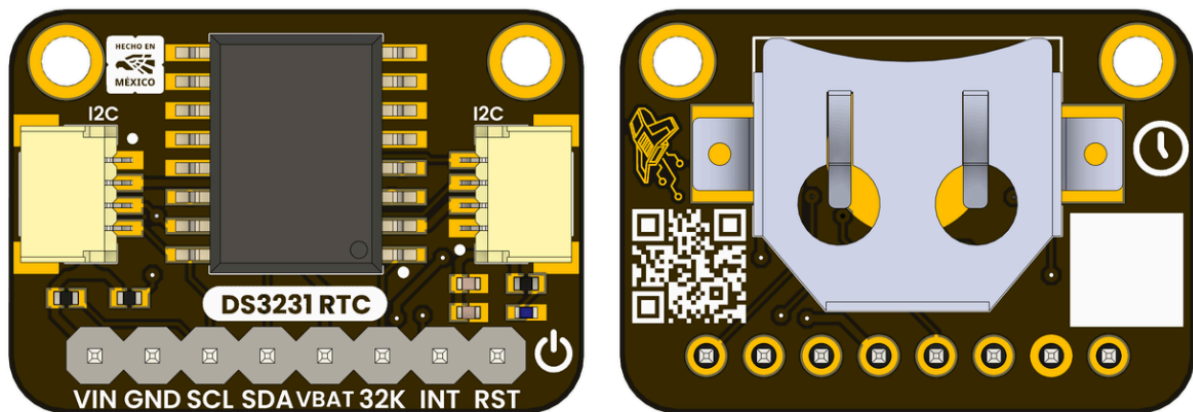
2 Ratings

2.1 Recommended Operating Conditions

Parameter	Description	Min	Typ	Max	Unit
Vcc	Input voltage to power on the module	2.3	3.3	5.5	V
Vbat	Battery Supply Voltage	2.3	3.0	5.5	V
Icc	Supply current	-	-	300	uA
Ilo	Output Leakage Current 32kHz, INT/SQW, SDA	-1	0	+1	uA
Ili	Input Leakage SCL	-1	-	+1	uA
Iol	RST Pin I/O Leakage	-200	-	+10	uA
Ibatlkg	VBAT Leakage Current (VCC Active)	-	25	100	nA
Vol	Logic 0 Output, 32kHz, INT/SQW, SDA	-	-	0.4V	V
Vol	Logic 0 Output, RST	-	-	0.4V	V
fout	Output Frequency	-	32.768	-	kHz
Ibata	Active Battery Current	-	-	150	uA
fscl	SCL Clock Frequency	0	-	400	KHz

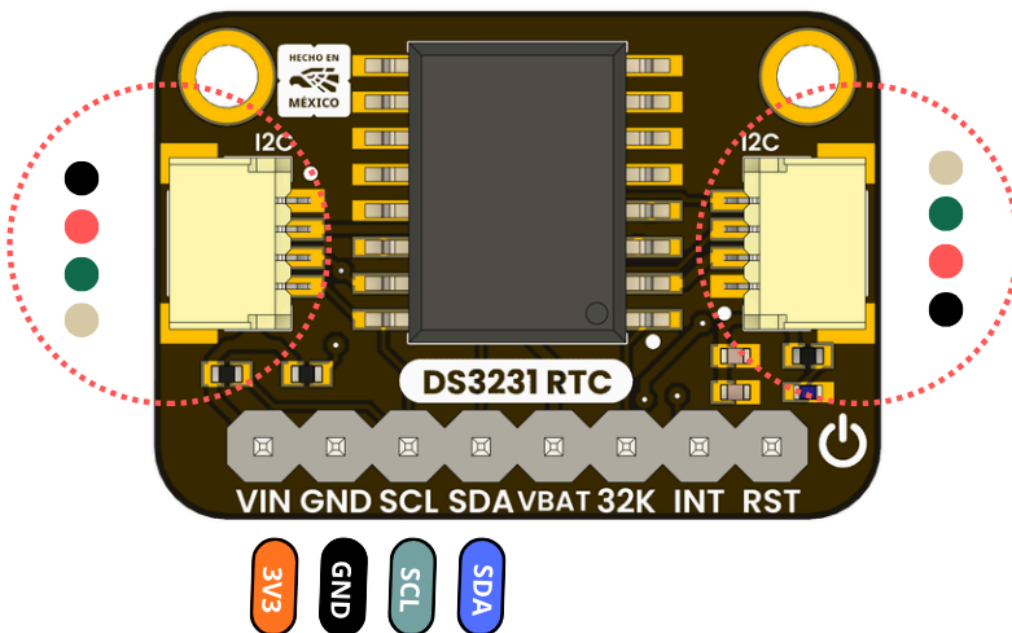
3 Functional Overview

The DS3231 RTC module integrates accurate timekeeping, hardware signaling, and flexible interfacing into a compact solution designed for embedded systems. Its functionality is organized across several key operational features described below.



3.1 I²C Communication Interface

The module operates as a slave device on a standard I²C bus, using a fixed 7-bit address of **0x68**. Communication is fully bidirectional, allowing the host microcontroller to read and write timekeeping registers, configuration settings, and status flags. All internal operations, including time updates, alarm configuration, and output control, are managed through this interface. The SDA and SCL lines are implemented as open-drain signals and require pull-up resistors, which are typically included on the module to simplify integration.



3.2 Timekeeping and Configuration

The device maintains complete calendar and time information, including seconds, minutes, hours, day, date, month, and year. Data is stored in binary-coded decimal (BCD) format, enabling efficient processing in embedded systems. The module supports both 12-hour and 24-hour formats, selectable via configuration registers. Automatic correction for months with fewer than 31 days and leap-year adjustment ensures long-term accuracy without external intervention. All parameters are configured through internal registers accessed over the I²C interface, allowing full software control of the device.

3.3 Alarm and Interrupt Operation

Two independent programmable alarms are integrated within the module, enabling time-based event generation. Each alarm can be configured to trigger under specific conditions, such as matching seconds, minutes, hours, day of the week, or calendar date. When an alarm condition is satisfied, the module asserts the INT/SQW output as an active-low signal. This pin operates in open-drain mode and can be used to generate hardware interrupts, initiate scheduled tasks, or wake the host system from low-power states. The interrupt functionality is controlled through dedicated configuration bits within the device registers.

3.4 Square-Wave and 32 kHz Output

The module provides flexible clock output capabilities through two independent signals. The INT/SQW pin can be configured to generate a programmable square-wave output at selectable frequencies, while a dedicated 32 kHz pin provides a continuous clock reference derived from the internal temperature-compensated oscillator. Both outputs are open-drain and require external pull-up resistors. These signals can be used for timing synchronization, clock distribution, or calibration of external subsystems.

3.5 Power Management and Backup Operation

The module supports dual power inputs, consisting of a primary supply (VCC) and a backup supply (VBAT). During normal operation, the device is powered from VCC. In the event of power loss, the system automatically switches to the VBAT input, typically connected to a coin-cell battery, ensuring uninterrupted timekeeping and retention of configuration data. This automatic switchover requires no software intervention and guarantees reliable operation in applications where power stability cannot be assured.

3.6 Hardware Connectivity and Integration

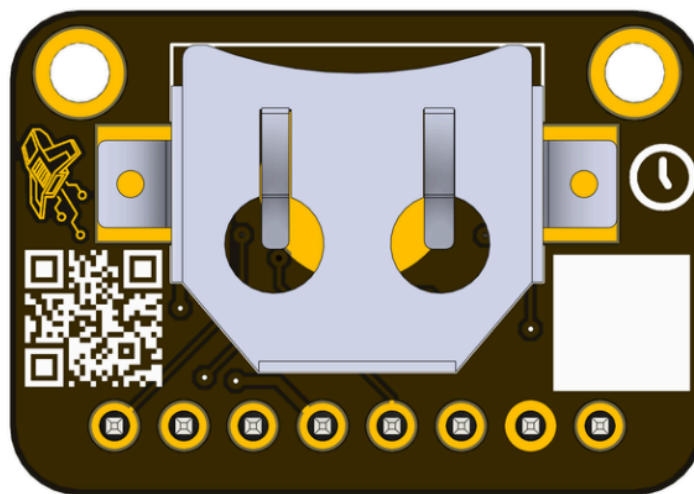
The module is designed to support both rapid prototyping and permanent integration. A JST 1.0 mm connector, commonly compatible with Qwiic and STEMMA QT ecosystems, provides a standardized interface for plug-and-play I²C communication. In parallel, the module exposes external header pins for direct connection to microcontrollers or custom PCBs. This dual connectivity approach enables flexible deployment across development environments.

and production systems. The inclusion of onboard pull-up resistors and clearly defined pin functions reduces the need for additional external components.

3.7 Battery Backup Connector

The DS3231 RTC module incorporates a dedicated battery backup connector designed to maintain uninterrupted timekeeping operation in the absence of the primary power supply. This connector is typically implemented as a coin-cell holder compatible with standard lithium batteries such as the **CR1220**, providing a stable auxiliary voltage source to the VBAT pin of the RTC device.

CR1220



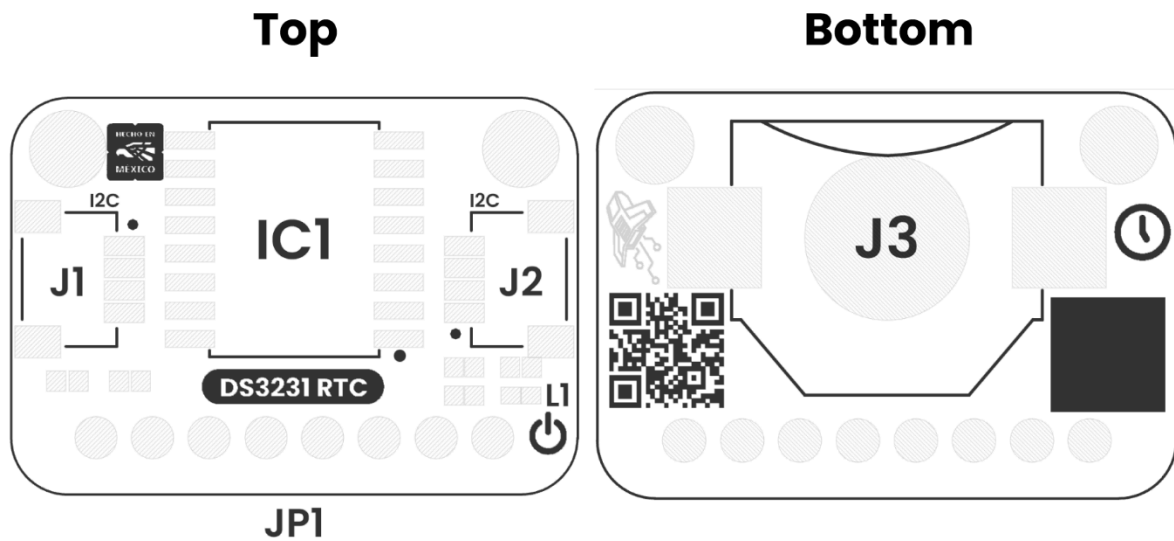
When the main supply voltage (VCC) is present, the module operates under normal conditions, and the backup battery remains electrically isolated from the primary power domain. In the event of a power loss, the internal power management circuitry of the DS3231 automatically switches to the VBAT input without requiring external control or software intervention. This seamless transition ensures that the internal oscillator continues running and that all timekeeping registers retain valid data.

The battery-backed operation mode is optimized for ultra-low power consumption, allowing the RTC to preserve accurate time over extended periods, typically ranging from several months to years depending on battery capacity and environmental conditions. During this mode, non-essential functions such as I²C communication are disabled, while core timekeeping functionality remains active.

The physical integration of the battery connector on the module simplifies system design by eliminating the need for external backup circuitry. It also ensures correct polarity and reliable electrical contact, reducing the risk of improper installation compared to discrete battery wiring solutions.

This feature is critical in applications such as data logging, industrial monitoring, and embedded control systems, where maintaining accurate time across power cycles is essential for system integrity and event traceability.

3.8 Board Topology



Views of Board Topology

Views of DRV2605L Haptic Motor Controller Module Topology

Table 3.2.1 - Components Overview

Ref.	Description
IC1	DS3231 RTC
L1	Power On LED
JP1	2.54 mm Header
J1	QWIIC Connector (JST 1 mm pitch) for I2C
J2	QWIIC Connector (JST 1 mm pitch) for I2C
J3	Battery Connector

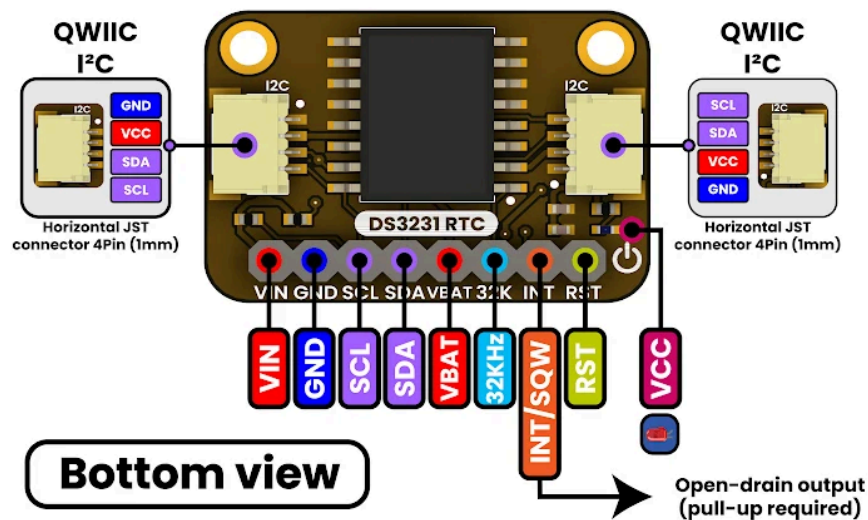
4 Connectors & Pinouts

4.1 General Pinout

PINOUT

DevLab: I2C DS3231 RTC Module

Top view



Bottom view



General Pinout

4.2 Pinout General Description

Pin Label	Function	Notes
VIN	Main power input	External supply input for the RTC module
GND	Ground reference	Common ground connection
SCL	I ² C clock line	I ² C serial clock signal
SDA	I ² C data line	I ² C serial data signal
VBAT	Backup battery input	Connects to CR1220 backup battery
32K	32.768 kHz clock output	Optional frequency reference output
INT/SQW	Interrupt / square wave output	Open-drain output, pull-up resistor required
RST	Reset input/output	Used for system reset monitoring
VCC	QWIIC connector power	Power line available on QWIIC I ² C connectors
QWIIC I ² C Connector	JST 4-pin I ² C interface	Horizontal JST 1 mm connector compatible with QWIIC ecosystem
BATTERY CLIP CR1220	Backup battery holder	Supports CR1220 coin cell battery (recommended 3V)

5 Board Operation

5.1 Getting Started with arduino IDE

1. Overview

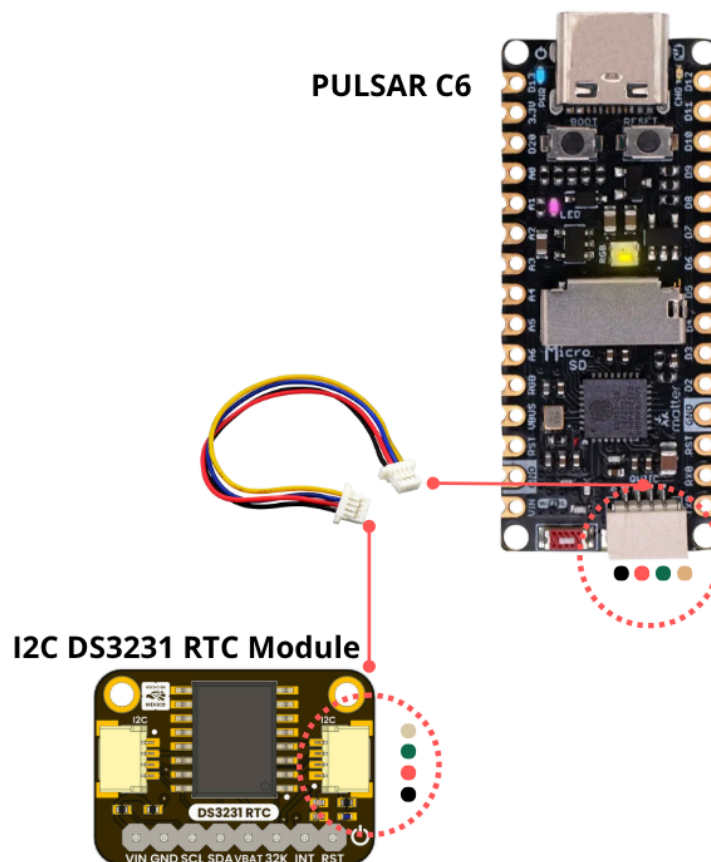
The DS3231 RTC is a high-accuracy real-time clock with an integrated temperature-compensated crystal oscillator (TCXO). It maintains time even when the main system is powered off by using a backup battery.

This guide demonstrates how to:

- Detect the RTC module
- Set the initial date and time
- Read and print the current time

2. Hardware Requirements

- Arduino-compatible board (ESP32, RP2040, AVR, etc.)
- DS3231 RTC module
- I²C connection (SDA, SCL)
- Backup battery (CR2032 recommended)



3. Wiring

Label	Description	Typical Connection (PULSAR C6)
VCC	Power supply	3.3V / 5V
GND	Ground	GND
SDA	I ² C Data	GPIO 6
SCL	I ² C Clock	GPIO 7

Note: On standard ESP32 boards, SDA/SCL are usually GPIO 21/22. Adjust as needed.

4. Library Installation

Install the following library in Arduino IDE:

- **RTClib** (by Adafruit)

5. Example Code

```
#include <Wire.h>
#include "RTClib.h"

RTC_DS3231 rtc;

#define SDA_PIN 6
#define SCL_PIN 7

const int SET_YEAR = 2026;
const int SET_MONTH = 5;
const int SET_DAY = 4;
const int SET_HOUR = 12;
const int SET_MINUTE = 0;
const int SET_SECOND = 0;
const bool FORCE_SET_TIME_ON_BOOT = false;

void printNow(const DateTime& now) {
  Serial.printf("%04d-%02d-%02d %02d:%02d:%02d\n",
```

```

        now.year(), now.month(), now.day(),
        now.hour(), now.minute(), now.second());
    }

void setup() {
    Serial.begin(115200);
    // Esperar USB estable (muy recomendado en ESP / RP)
    uint32_t t0 = millis();
    while (!Serial && millis() - t0 < 2000) {
        delay(10);
    }
    delay(200);
    Wire.begin(SDA_PIN, SCL_PIN);
    if (!rtc.begin()) {
        Serial.println("ERROR: DS3231 not found");
        while (1) delay(10);
    }
    if (FORCE_SET_TIME_ON_BOOT || rtc.lostPower()) {
        rtc.adjust(DateTime(SET_YEAR, SET_MONTH, SET_DAY,
                            SET_HOUR, SET_MINUTE, SET_SECOND));
        Serial.println("RTC initialized");
    }
    Serial.print("RTC OK: ");
    printNow(rtc.now());
}

void loop() {
    static unsigned long lastPrint = 0;

```

```
if (millis() - lastPrint >= 1000) {  
  
    Serial.print("[RTC] ");  
  
    printNow(rtc.now());  
  
    lastPrint = millis();  
  
}  
  
}
```

6. Expected Output

Open the Serial Monitor (115200 baud):

```
RTC initialized  
RTC OK: 2026-05-04 12:00:00  
[RTC] 2026-05-04 12:00:01  
[RTC] 2026-05-04 12:00:02
```

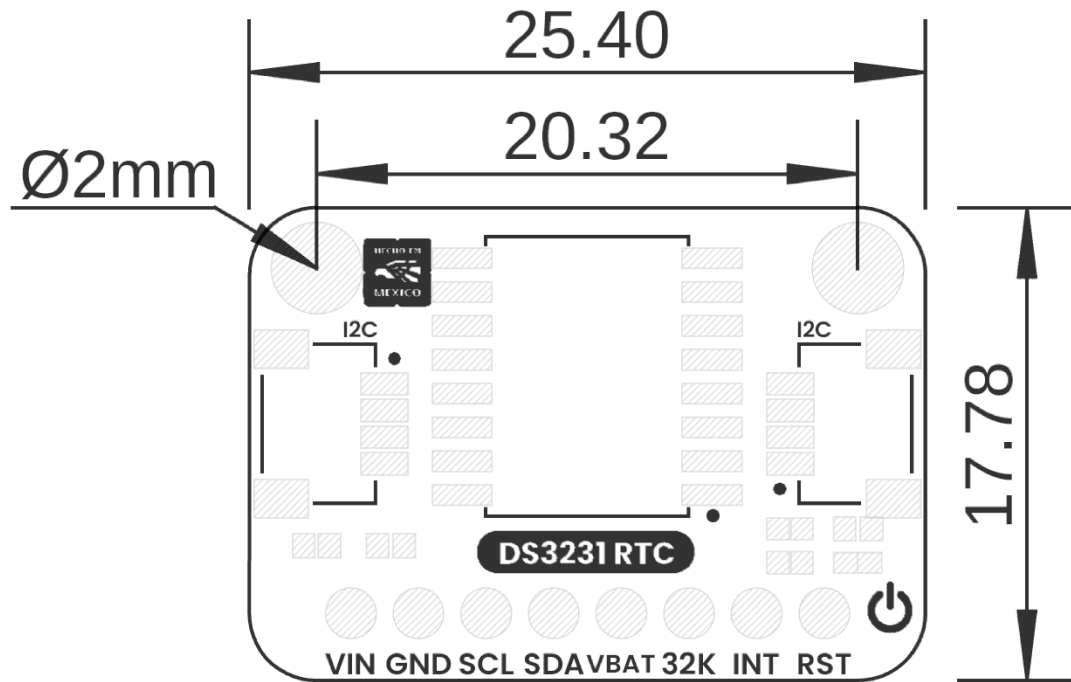
7. Important Notes

- **First Run Only:**
Set `FORCE_SET_TIME_ON_BOOT = true` to initialize the RTC.
Then change it to `false` to preserve time after reset.
- **Battery Backup:**
Ensure a CR2032 battery is installed; otherwise, time will reset after power loss.
- **I²C Address:**
Default address is `0x68`.

8. Troubleshooting

Issue	Possible Cause	Solution
RTC not detected	Wiring error	Check SDA/SCL connections
Time resets	No battery	Install CR2032
No Serial output	Baud mismatch	Set Serial Monitor to 115200

6 Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	Link
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_ds3231_rtc_module
Wiki	https://wiki.uelectronics.com/wiki/unit_devlab_i2c_ds3231_rtc_module
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1

Schematic(https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_ds3231_rtc_module/blob/main/hardware/unit_sch_v_1_0_0_ue0107_ds3231_rtc_module.pdf)

